UNIVERSAL INJECTIVITY APPEARS AT A FINITE STAGE IN THE DESCENT SITUATION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1835-1846, following the statement at 1825-1833.

Editorial note. No omissions.

Allowed standard prerequisites (editorial).

- Universally injective via surjective diagonal. Morphisms of Spaces, Definition spaces-morphisms-definition-universally-injective and Lemma spaces-morphisms-lemma-universally-injective (cross-chapter, cited at ~1835-1840). Universally injective iff diagonal X -> X x_Y X surjective.
- Diagonal is locally of finite presentation. Morphisms of Spaces, Lemma spaces-morphisms-lemma-diagonal-morphism-finite-type (cross-chapter, cited at ~1840-1843). X_0 -> X_0 x_{Y_0} X_0 is lfp.
- Surjectivity appears at a finite stage. lemma-descend-surjective (same chapter, card L26329, cited at ~1844-1846). Applied to the diagonal X_0 -> X_0 x_{Y_0} X_0.

Source reference. Original lemma 07SN: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1776–1776. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 084W: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1666–1666. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03MV: focused-seeds/geometry/S9-spaces-limits-descend-universally-injective/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 084P: focused-seeds/geometry/S9-spaces-limits-descend-universally-injective/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 040X: focused-seeds/geometry/S9-spaces-limits-descend-universally-injective/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Notation and assumptions as in Situation 084W. If*

(1) *f is universally injective,*
(2) *f_0 is locally of finite type,*

then f_i is universally injective for some $i \geq 0$.

Proof. Recall that a morphism $X \to Y$ is universally injective if and only if the diagonal $X \to X \times_Y X$ is surjective (Morphisms of Spaces, Definition 03MV and Lemma 040X). Observe that $X_0 \to X_0 \times_{Y_0} X_0$ is of locally of finite presentation (Morphisms of Spaces, Lemma 084P). Hence the lemma follows from Lemma 07SN by considering the morphism $X_0 \to X_0 \times_{Y_0} X_0$. □